6.6 THE BUCK-BOOST CONVERTER

Another basic switched-mode converter is the buck-boost converter shown in Fig. 6-11. The output voltage of the buck-boost converter can be either higher or lower than the input voltage.

Voltage and Current Relationships

Assumptions made about the operation of the converter are as follows:

1. The circuit is operating in the steady state.
2. The inductor current is continuous.
3. The capacitor is large enough to assume a constant output voltage.
4. The switch is closed for time DT and open for $(1-D)T$.
5. The components are ideal.

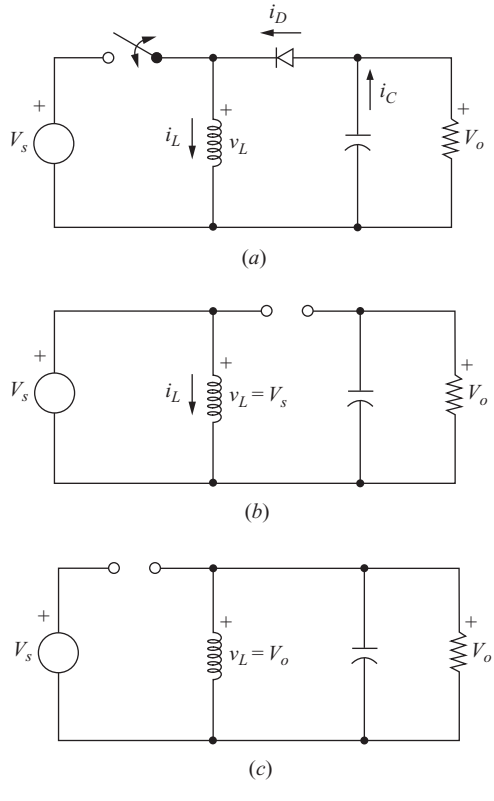


Figure 6-11 Buck-boost converter. (*a*) Circuit; (*b*) Equivalent circuit for the switch closed; (*c*) Equivalent circuit for the switch open.

Analysis for the Switch Closed When the switch is closed, the voltage across the inductor is

$$v_L = V_s = L\frac{di_L}{dt}$$

$$\frac{di_L}{dt} = \frac{V_s}{L}$$

The rate of change of inductor current is a constant, indicating a linearly increasing inductor current. The preceding equation can be expressed as

$$\frac{\Delta i_L}{\Delta t} = \frac{\Delta i_L}{DT} = \frac{V_s}{L}$$

Solving for Δi_L when the switch is closed gives

$$(\Delta i_L)_{\text{closed}} = \frac{V_s DT}{L} \tag{6-45}$$

Analysis for the Switch Open When the switch is open, the current in the inductor cannot change instantaneously, resulting in a forward-biased diode and current into the resistor and capacitor. In this condition, the voltage across the inductor is

$$v_L = V_o = L\frac{di_L}{dt}$$

$$\frac{di_L}{dt} = \frac{V_o}{L}$$

Again, the rate of change of inductor current is constant, and the change in current is

$$\frac{\Delta i_L}{\Delta t} = \frac{\Delta i_L}{(1-D)T} = \frac{V_o}{L}$$

Solving for Δi_L,

$$(\Delta i_L)_{\text{open}} = \frac{V_o(1-D)T}{L} \tag{6-46}$$

For steady-state operation, the net change in inductor current must be zero over one period. Using Eqs. (6-45) and (6-46),

$$(\Delta i_L)_{\text{closed}} + (\Delta i_L)_{\text{open}} = 0$$

$$\frac{V_s DT}{L} + \frac{V_o(1-D)T}{L} = 0$$

Solving for V_o,

$$\boxed{V_o = -V_s\left(\frac{D}{1-D}\right)} \tag{6-47}$$

The required duty ratio for specified input and output voltages can be expressed as

$$D = \frac{|V_o|}{V_s + |V_o|} \tag{6-48}$$

The average inductor voltage is zero for periodic operation, resulting in

$$V_L = V_s D + V_o(1 - D) = 0$$

Solving for V_o yields the same result as Eq. (6-47).

Equation (6-47) shows that the output voltage has opposite polarity from the source voltage. *Output voltage magnitude of the buck-boost converter can be less than that of the source or greater than the source, depending on the duty ratio of the switch.* If $D > 0.5$, the output voltage is larger than the input; and if $D < 0.5$, the output is smaller than the input. Therefore, this circuit combines the capabilities of the buck and boost converters. Polarity reversal on the output may be a disadvantage in some applications, however. Voltage and current waveforms are shown in Fig. 6-12.

Note that the source is never connected directly to the load in the buck-boost converter. Energy is stored in the inductor when the switch is closed and transferred to the load when the switch is open. Hence, the buck-boost converter is also referred to as an *indirect* converter.

Power absorbed by the load must be the same as that supplied by the source, where

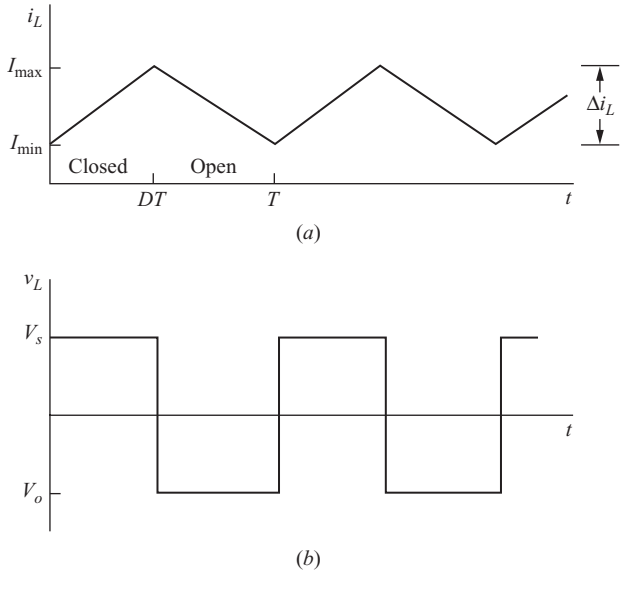


Figure 6-12 Buck-boost converter waveforms. (*a*) Inductor current; (*b*) Inductor voltage; (*c*) Diode current; (*d*) Capacitor current.

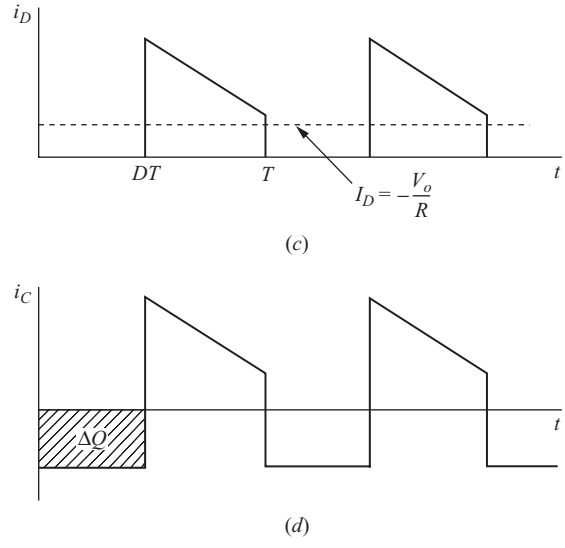


Figure 6-12 *(continued)*

$$P_o = \frac{V_o^2}{R}$$

$$P_s = V_s I_s$$

$$\frac{V_o^2}{R} = V_s I_s$$

Average source current is related to average inductor current by

$$I_s = I_L D$$

resulting in

$$\frac{V_o^2}{R} = V_s I_L D$$

Substituting for V_o using Eq. (6-47) and solving for I_L, we find

$$I_L = \frac{V_o^2}{V_s R D} = \frac{P_o}{V_s D} = \frac{V_s D}{R(1-D)^2} \tag{6-49}$$

Maximum and minimum inductor currents are determined using Eqs. (6-45) and (6-49).

$$I_{\max} = I_L + \frac{\Delta i_L}{2} = \frac{V_s D}{R(1-D)^2} + \frac{V_s D T}{2L} \tag{6-50}$$

$$I_{\min} = I_L - \frac{\Delta i_L}{2} = \frac{V_s D}{R(1-D)^2} - \frac{V_s D T}{2L} \tag{6-51}$$

For continuous current, the inductor current must remain positive. To determine the boundary between continuous and discontinuous current, $I_{\min}$ is set to zero in Eq. (6-51), resulting in

$$(Lf)_{\min} = \frac{(1-D)^2 R}{2} \tag{6-52}$$

or

$$\boxed{L_{\min} = \frac{(1-D)^2 R}{2f}} \tag{6-53}$$

where f is the switching frequency.

Output Voltage Ripple

The output voltage ripple for the buck-boost converter is computed from the capacitor current waveform of Fig. 6-12*d*.

$$|\Delta Q| = \left(\frac{V_o}{R}\right) DT = C\Delta V_o$$

Solving for ΔV_o,

$$\Delta V_o = \frac{V_o DT}{RC} = \frac{V_o D}{RCf}$$

or

$$\boxed{\frac{\Delta V_o}{V_o} = \frac{D}{RCf}} \tag{6-54}$$

As is the case with other converters, the equivalent series resistance of the capacitor can contribute significantly to the output ripple voltage. The peak-to-peak variation in capacitor current is the same as the maximum inductor current. Using the capacitor model shown in Fig. 6-6, where $I_{L,\max}$ is determined from Eq. (6-50),

$$\Delta V_{o,\text{ESR}} = \Delta i_C r_C = I_{L,\max} r_C \tag{6-55}$$

EXAMPLE 6-6

Buck-Boost Converter

The buck-boost circuit of Fig. 6-11 has these parameters:

$V_s = 24$ V
$D = 0.4$
$R = 5\ \Omega$
$L = 20\ \mu$H
$C = 80\ \mu$F
$f = 100$ kHz

Determine the output voltage, inductor current average, maximum and minimum values, and the output voltage ripple.

■ **Solution**

Output voltage is determined from Eq. (6-47).

$$V_o = -V_s\left(\frac{D}{1-D}\right) = -24\left(\frac{0.4}{1-0.4}\right) = -16 \text{ V}$$

Inductor current is described by Eqs. (6-49) to (6-51).

$$I_L = \frac{V_s D}{R(1-D)^2} = \frac{24(0.4)}{5(1-0.4)^2} = 5.33 \text{ A}$$

$$\Delta i_L = \frac{V_s DT}{L} = \frac{24(0.4)}{20(10)^{-6}(100{,}000)} = 4.8 \text{ A}$$

$$I_{L,\max} = I_L + \frac{\Delta i_L}{2} = 5.33 + \frac{4.8}{2} = 7.33 \text{ A}$$

$$I_{L,\min} = I_L - \frac{\Delta i_L}{2} = 5.33 - \frac{4.8}{2} = 2.93 \text{ A}$$

Continuous current is verified by $I_{\min} > 0$.

Output voltage ripple is determined from Eq. (6-54).

$$\frac{\Delta V_o}{V_o} = \frac{D}{RCf} = \frac{0.4}{(5)(80)(10)^{-6}(100{,}000)} = 0.01 = 1\%$$